Dexter Wen Jie Teo

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RESEARCH INTERESTS

Medical Image Analysis, Semi-Supervised Learning, Image Segmentation, Spatiotemporal Modeling, Multimodal Data Fusion, Clinical Robustness & Generalizability

EDUCATION

Nanyang Technological University (NTU), Singapore Aug 2023 – Present
Bachelor of Engineering (Honours), Computer Engineering

- Expected Grade: Highest Distinction | Current cGPA: 4.80/5.00
- Elective Focus in Artificial Intelligence; Elective Focus in Cybersecurity
- Dean's List (Top 5%) for Academic Year 2023/24, 2024/25; CJ Koh Scholarship
- Relevant Coursework: Computer Vision; Linear Algebra for Computing; Data Structures & Algorithms; Embedded Programming

University of Waterloo (UWaterloo), Canada Aug 2025 – Dec 2025
GEM Explorer, Student Exchange Program

- Relevant Coursework: Foundations of Artificial Intelligence; Introduction to Machine Learning; Signals and Systems

Singapore Management University (SMU), Singapore Jan 2025 – May 2025
Singapore Universities Student Exchange Programme

- Term GPA: 4.0/4.0
- Relevant Coursework: IT Solution Architecture; Design & Analysis of Algorithms

PUBLICATIONS

- Shehata, N., **Teo, D. W. J.**, Dell'Aniello Picard, F., Poudel, P., Yacoub, M. H., Lombaert, H. "Single-slice 3D aortic reconstruction using implicit neural representations." *SPIE Medical Imaging 2027: Image Processing*, paper 14518-67, accepted for presentation, 2027.
- Shehata, N., **Teo, D. W. J.**, Dell'Aniello Picard, F., Poudel, P., Yacoub, M., Lombaert, H. "3D Shape Reconstruction from a Single 2D Slice Using Implicit Neural Representations: Application to the Aorta." *Medical Image Analysis (Under Review, 2026).*
- **Teo, D. W. J.**, Shehata, N., Lombaert, H. "Spatiotemporal Distillation via Recurrent Bottlenecks for Aortic Tracking". *MICCAI Workshop on Statistical Atlases and Computational Modeling of the Heart (STACOM),* 2026.
- **Teo, D. W. J.**, Punithakumar, K. "Native-Space 3D CarveMix for Multi-Site T1w Stroke Segmentation". *MICCAI Stroke and neurovascular diseases Workshop on Imaging and Treatment CHallenges (SWITCH+),* 2026.
- Guegan, T. and **Teo, D. W. J.** "Behavior Cloning of MPC for 3-DOF Robotic Manipulators." *ICRA Workshop on Reinforcement Learning in the Era of Imitation Learning (RL4IL)*, 2026.

RESEARCH EXPERIENCE

Nanyang Technological University, Singapore Aug 2026 – Present
Final Year Project *(Supervisor: Prof. Qian Kemao)*

- Extending my University of Alberta research for ISLES 2026 on native-space stroke lesion segmentation in T1-weighted MRI
- Investigating data augmentation, loss functions, and multiscale deep supervision to improve small-lesion detection sensitivity and boundary accuracy

University of Alberta, Canada Jun 2026 – Aug 2026

Mitacs Globalink Research Intern *(Supervisor: Prof. K. Punithakumar)*
- Awarded a competitive, fully funded research fellowship
- Engineered an on-the-fly 3D CarveMix augmentation engine that dynamically synthesized acute stroke lesion placements directly into native T1w MRI data
- Evaluated 3D MedNeXt architectures across 1,453 multi-site scans, achieving a +0.018 Dice score gain across acute, subacute, and chronic cohorts
- Ranked 4th in the ISLES 2026 Challenge at MICCAI 2026

Polytechnique Montréal, Canada Jan 2026 – May 2026
Research Intern *(Supervisor: Prof. Hervé Lombaert)*
- Selected as one of 25 students out of 250 annually for PolyMTL Research Scholarship
- Developed a semi-supervised spatiotemporal distillation framework in PyTorch/MONAI for aortic segmentation and tracking in cardiac cine-MRI
- Validated on 437 patient scans, achieving 92.3% normalized surface Dice (NSD) at a 1 mm tolerance and reducing sequence-wide structural anomalies by over 56% compared to a 2D nnU-Net baseline

PROJECT WORK

Machine Learning Project Aug 2025 – Dec 2025
Multimodal Cardiomegaly Detection from Chest X-rays with Demographics *(Grade: 95% – UWaterloo)*
- **Role: Lead Architect & Evaluator (Team of 3):** Designed and validated multimodal deep learning frameworks using the MIMIC-CXR dataset
- **Architectural Exploration:** Conducted an extensive comparative study across 6+ architectures – Logistic Regression, Random Forest, Gradient Boosting, SVM, and custom Attention-based networks
- **Model Evaluation:** Improved ROC-AUC from 0.79 to 0.83 and boosted sensitivity by ~17%, demonstrating that advanced fusion techniques significantly outperformed traditional baselines
- **Clinical Insights:** Performed feature-importance analysis showing age as the primary demographic driver; documented limitations and future extensions

SKILLS

- Programming: Python (PyTorch, MONAI), C, Java
- Languages: English, Mandarin Chinese

CERTIFICATIONS

- Introduction to Digital Health, Imperial College London
- AWS Certified Solutions Architect – Associate
- CS50 Introduction to Cybersecurity, Harvard University
- CS50x Introduction to Computer Science, Harvard University

WORK EXPERIENCE

DBS Bank Singapore May 2025 – Aug 2025
Technology Risk Management Intern
- Participated in Targeted Risk Reviews regarding Infrastructure Change Management
- Assisted in creation of Bank's Standard Operating Procedures on Handling Disciplinary Action Cases

National Service, Singapore Armed Forces Jan 2021 – Dec 2022
3rd Sergeant, Infantry
- Appointed as a Third Sergeant in 5th Singapore Infantry Regiment (5 SIR)
- Ensured operational readiness of a fleet of 12 Terrex Infantry Carrier Vehicles (ICVs)